LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 4 Test (GSEB pattern): Carbon and its Compounds

Chapter / Topic: Carbon and its Compounds

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. Ethane, with the molecular formula C_2H_6, has: **[1]**

(a) 6 covalent bonds (b) 7 covalent bonds
(c) 8 covalent bonds (d) 9 covalent bonds

Q2. The next homologue of methane (CH_4) is: **[1]**

(a) C_2H_6 (b) C_2H_4
(c) C_3H_8 (d) CH_3OH

Q3. Butanone is a four-carbon compound with the functional group: **[1]**

(a) carboxylic acid (b) aldehyde
(c) ketone (d) alcohol

Q4. The functional group present in propanal is: **[1]**

(a) alcohol (b) ketone
(c) carboxylic acid (d) aldehyde

Q5. The next higher homologue of ethanol (C_2H_5OH) is: **[1]**

(a) CH_3OH (b) C_3H_7OH
(c) C_2H_5COOH (d) C_3H_8

Q6. The two carbon atoms in ethene are joined by: **[1]**

(a) a single bond (b) a triple bond
(c) a double bond (d) an ionic bond

Q7. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that: **[1]**

(a) the fuel is not burning completely
(b) the food is not cooked completely
(c) the fuel is wet
(d) the fuel is burning completely

Q8. Soaps are: **[1]**

(a) esters of ethanol
(b) ammonium salts of short-chain acids
(c) alcohols
(d) sodium or potassium salts of long-chain carboxylic acids

Q9. Which of the following is an unsaturated hydrocarbon? **[1]**

(a) C_3H_8 (b) C_2H_4
(c) CH_4 (d) C_2H_6

SECTION B

6 marks

Q10. What is a homologous series? Write the formulae of the first two members of the alkyne series. **[2]**

Q11. Why do covalent compounds generally have low melting and boiling points? Why are they poor conductors of electricity? **[2]**

Q12. What is saponification? Write the balanced equation for the reaction of ethyl ethanoate with sodium hydroxide. **[2]**

SECTION C

6 marks

Q13. Answer the following about ethanol: **[3]**

(i) Write the balanced equation for the reaction of ethanol with sodium. **[1]**

(ii) What is formed when ethanol is heated with excess concentrated sulphuric acid at 443 K? Write the equation. **[1]**

(iii) Name the product formed when ethanol is oxidised with alkaline potassium permanganate. **[1]**

Q14. Answer the following: **[3]**

(i) Write the general formula of alkenes. **[1]**

(ii) Write the name and formula of the third member of the alkane series. **[1]**

(iii) By how much do the molecular masses of two successive members of a homologous series differ? **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. Carbon compounds burn in oxygen to give carbon dioxide and water, along with heat and light. Saturated hydrocarbons generally burn with a clean blue flame, while unsaturated hydrocarbons give a yellow flame with lots of black smoke. In a gas stove, the air holes let in enough oxygen for complete combustion. If the air holes get blocked, the flame becomes yellow and sooty. **[4]**

(i) Write the balanced equation for the complete combustion of methane. **[1]**

(ii) Which type of hydrocarbon burns with a sooty yellow flame? **[1]**

(iii) Why does the flame become sooty when the air holes are blocked, and why is this a problem? **[2]**

~ All the best! ~